Tutorial 3

MAT3007 Optimization, 2022 Fall

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Outline

1. Graphical Solutions to LP
2. Polyhedron
3. Convex and Extreme Points
4. BS, BFS

Exercise 1: Graphical Solutions to LP

Consider the follow problem:

$$\begin{aligned} \max \quad & 15x_1 + 10x_2 \\ \text{s.t.} \quad & 4x_1 + 6x_2 \leq 360 \\ & 3x_1 \leq 180 \\ & 5x_2 \leq 200 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Steps to solve 2-D LP graphicly:

1. Draw the feasible region
2. Draw the function $15x_1 + 10x_2 = c$.

Feasible Region

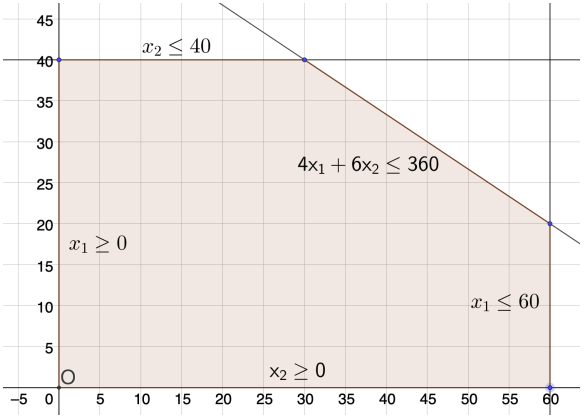
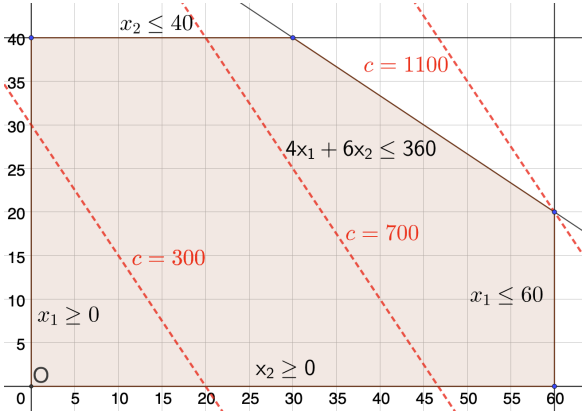


Figure: Feasible Region

Different Objectives

Draw the lines which represent $15x_1 + 10x_2 = c$:



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Definition Review

Definition 1 (Polyhedron).

- Definition:
  - $S = \{x \in R^n \mid Ax \geq b\}$
  - The **intersection** of **finite** number of **linear** equalities and inequalities.
- Standard form (**a special case**): $S = \{x \in R^n \mid Ax = b, x \geq 0\}$

Exercise 2: Polyhedron

As we all know, the feasible set of LP is a polyhedron. For each one of the following sets, determine whether it is a Polyhedron

(a) the set of all $(x, y) \in R^2$ satisfying the constraints.

$$s.t. \quad x\cos\theta + y\sin\theta \leq 1 \qquad \forall\theta \in [0, \pi/2]$$
$$x, y \geq 0$$

(b) $S = \{(x, y) \in R^2 \mid x^2 - 8x + 15 \leq y\}$

(c) $\emptyset$

(d) $S = \left\{x \in R^n \mid x \geq 0, x^T y \leq 1, \textit{for all } y \textit{ with } \|y\|_2 = 1\right\}$

(e) $S = \left\{x \in R^n \mid x \geq 0, x^T y \leq 1, \textit{for all } y \textit{ with } \|y\|_1 = 1\right\}$

Solution to Exercise 2

Solution:

(a) $\times$ infinite inequalities.

(b) $\times$ nonlinear inequalities.

(c) $\surd$ $\emptyset = \{x \mid x \geq 0\} \cap \{x \mid x < 0\}$

(d) $\times$ $S = \left\{x \in R^n \mid x \geq 0, x^T y \leq 1, \textit{for all } y \textit{ with } \|y\|_2 = 1\right\}$
$= \{x \in R^n \mid x \geq 0, \|x\|_2 \leq 1\}$

(e) $\surd$ $S = \left\{x \in R^n \mid x \geq 0, x^T y \leq 1, \textit{for all } y \textit{ with } \|y\|_1 = 1\right\}$
$= \{x \in R^n \mid x \geq 0, \|x\|_\infty \leq 1\}$

Solution to Exercise 2

Proof of (d):

- set $S_1 = \left\{ x \in R^n \mid x \geq 0, x^T y \leq 1, \textit{for all } y \textit{ with } \|y\|_2 = 1 \right\}$
  $S_2 = \{x \in R^n \mid x \geq 0, \|x\|_2 \leq 1\}$
  In order to prove $S_1 = S_2$, we only need to prove $S_2 \subset S_1$ and $S_1 \subset S_2$
- $\Leftarrow \forall x \in S_2$ *with* $\|x\|_2 \leq 1$, $\forall y \in R^n$ *with* $\|y\|_2 = 1$,
  By Cauchy-Schwartz inequality,
  $x^T y \leq \|x\|_2 \|y\|_2 \leq 1$,
  so $x \in S_1$, $S_2 \subset S_1$
- $\Rightarrow \forall x \in S_1$, if $x = 0$, then obviously $x \in S_2$,
  if $x \neq 0$, choose a special $y = \frac{x}{\|x\|_2}$ satisfying $\|y\|_2 = 1$
  then $1 \geq x^T y = \frac{x^T x}{\|x\|_2} = \|x\|_2$
  so $x \in S_2$, $S_1 \subset S_2$

Solution to Exercise 2

Proof of (e):

(d) S is a polyhedron. S is the intersection of the set $\{x \mid |x_k| \leq 1, \ \ k = 1, \ldots, n\}$ and the nonnegative orthant $\mathbf{R}^n_+$. This follows from the following fact:

$$x^T y \leq 1 \text{ for all } y \text{ with } \sum_{i=1}^{n} |y_i| = 1 \iff |x_i| \leq 1, \quad i = 1, \ldots, n.$$

We can prove this as follows. First suppose that $|x_i| \leq 1$ for all i. Then

$$x^T y = \sum_i x_i y_i \leq \sum_i |x_i||y_i| \leq \sum_i |y_i| = 1$$

if $\sum_i |y_i| = 1$.

Conversely, suppose that x is a nonzero vector that satisfies $x^T y \leq 1$ for all y with $\sum_i |y_i| = 1$. In particular we can make the following choice for y: let k be an index for which $|x_k| = \max_i |x_i|$, and take $y_k = 1$ if $x_k > 0$, $y_k = -1$ if $x_k < 0$, and $y_i = 0$ for $i \neq k$. With this choice of y we have

$$x^T y = \sum_i x_i y_i = y_k x_k = |x_k| = \max_i |x_i|.$$

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Definition Review

Definition 2 (Convex Sets).

A set $S \subseteq \mathbb{R}^n$ is convex if for any x, $y \in S$, and any $\lambda \in [0,1]$, $\lambda x + (1-\lambda)y \in S$.

Definition 3 (Convex Combination).

For any $x_1, ..., x_n$ and $\lambda_1, ..., \lambda_n \geq 0$ satisfying $\lambda_1 + \cdots + \lambda_n = 1$, we call $\sum_{i=1}^{n} \lambda_i x_i$ a convex combination of $x_1, ..., x_n$.

Definition 4 (Extreme Point).

Let P be a polyhedron. A point $x \in P$ is said to be an extreme point of P if we can not find two elements $y, z \in P$ with $y, z \neq x$ and a scalar $\lambda \in [0,1]$, such that $x = \lambda y + (1-\lambda)z$.

Exercise 3

Let $S \in \mathbb{R}^n$ and $S^* \in \mathbb{R}^m$ be two convex sets. Suppose $T \in \mathbb{R}^{m \times n}$ is matrix and T is a bijective mapping from S to S^*. Show that T is also a bijective mapping from the extreme points of S to the extreme points of S^*.

Note: $T : S \to S^*$ is called bijective if

- T is surjective, i.e., $T(S) = S^*$.
- T is injective, i.e., $\forall x \neq y \in S, T(x) \neq T(y)$.

Solution to Exercise 3 I

Since $T : S \to S^*$ is a bijective mapping, we can define the inverse mapping of T, $T^{-1} : S^* \to S$. We claim T^{-1} and T are both linear. T is a matrix so it is linear. For T^{-1}, notice that:

$$T(\alpha T^{-1}(x) + \beta T^{-1}(y)) = \alpha T(T^{-1}(x)) + \beta T(T^{-1}(y)) = \alpha x + \beta y$$

which means:

$$T^{-1}(\alpha x + \beta y) = \alpha T^{-1}(x) + \beta T^{-1}(y)$$

So T^{-1} is also linear.
Let E and E^* be the set consisting of extreme points of S and S^* respectively. To prove the conclusion, it suffices to show that $T(E) \subset E^*$ and $T^{-1}(E^*) \subset E$.

Solution to Exercise 3 II

- $T(E) \subset E^*$. Assume that there is a point $x \in E$ but $T(x) \notin E^*$. By the definition of extreme point, there are two points $y, z \in S^*$ such that $y \neq T(x), z \neq T(x)$ and $\alpha y + (1-\alpha)z = T(x)$ with $\alpha \in [0,1]$. But since T^{-1} is linear, we can deduce that:

$$x = T^{-1}(T(x)) = T^{-1}(\alpha y + (1-\alpha)z) = \alpha T^{-1}(y) + (1-\alpha)T^{-1}(z).$$

  Moreover, since T^{-1} is injective, the condition that $y, z \neq T(x)$ implies $T^{-1}(y), T^{-1}(z) \neq x$. So x is not an extreme point, which contradicts the assumption $x \in E$. This shows that $T(E) \subset E^*$.
- $T^{-1}(E^*) \subset E$. The proof is similar because T is also injective and linear.

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Definition Review

Definition 5 (Basic Solution).

We call x a basic solution of the LP if and only if

1. $Ax = b$.
2. There exist indices $B(1), ..., B(m)$ such that the columns of

$$\begin{bmatrix} | & | & & | \\ A_{B(1)} & A_{B(2)} & \cdots & A_{B(m)} \\ | & | & & | \end{bmatrix}$$

are linearly independent and $x_i = 0$ for $i \neq B(1), ..., B(m)$.

Definition 6 (Basic Feasible Solutions).

If a basic solution x also satisfies that $x \geq 0$, then we call it a basic feasible solution (BFS).

Exercise 4: Fundamental LP Theorem

Theorem 7.

1. *Suppose the feasible set is* $\{x \in \mathbb{R}^n | Ax = b, x \geq 0\}$*, where* $A \in \mathbb{R}^{m \times n}$ *and* $b \in \mathbb{R}^m$*. If it is nonempty, then there is a basic feasible solution.*
2. *If there is an optimal solution, there is an optimal solution that is also a basic feasible solution.*

Lemma

Lemma 8.

Suppose there is a feasible solution x. Let $B = \{i | x_i > 0\}$ and $k = |B|$. If x is not a basic solution, then there exists $\alpha \in \mathbb{R}^n$ and an index $j \in B$ such that $\alpha_j > 0$ and $A\alpha = 0$.

- Denote the columns of A by $A_1, A_2, \ldots, A_n$.
- Since x is not a basic solution, $A_{B(1)}, A_{B(2)}, \ldots, A_{B(k)}$ are linear dependent.
- There exist α_i such that $\sum_{i \in B} \alpha_i A_i = 0$ with ar least one α_j is positive. Define $\alpha \in \mathbb{R}^n$ by

$$\alpha = \begin{cases} \alpha_i & i \in B \\ 0 & \text{otherwise} \end{cases}$$

Then $A\alpha = \sum_{i \in B} \alpha_i A_i = 0$.

Proof of (1)

- If x is a feasible point but not a basic solution, by lemma, there exist $\alpha \in \mathbb{R}^n$ with $A\alpha = 0$ and $\alpha_j > 0$, where j is an index in B.
- Since $x_i > 0$ for $i \in B$ and $\alpha_j > 0$, there exist an ϵ such that $x - \epsilon\alpha \geq 0$ and there is one more 0 entry in it.
- $A(x - \epsilon\alpha) = Ax - \epsilon A\alpha = b$, so $\bar{x} = x - \epsilon\alpha$ is still feasible.
- We can do this operation once and once again until we obtain a basic solution.

Proof of (2)

- If x is an optimal solution but not an optimal basic feasible solution, by lemma, there exist $\alpha \in \mathbb{R}^n$ with $A\alpha = 0$ and $\alpha_j > 0$, where j is an index in B.
- Since $x_i > 0$ for $i \in B$ and $\alpha_j > 0$, there exist an ϵ such that $x - \epsilon\alpha \geq 0$ and there is one more 0 entry in it.
- $A(x - \epsilon\alpha) = Ax - \epsilon A\alpha = b$, so $\bar{x} = x - \epsilon\alpha$ is still feasible.
- If $c^\top \alpha \neq 0$, then we can construct a better solution than x by considering $x + \epsilon\alpha$ or $x - \epsilon\alpha$, which contradicts with the optimality of x. So $c^\top \alpha$, and $c^\top \bar{x} = c^\top x + \epsilon c^\top \alpha = c^\top x$, $\bar{x}$ is still an optimal solution.
- We can do this operation once and once again, until obtain an optimal basic feasible solution.

Example

$$
\begin{aligned}
\min \quad & -3x_1 - x_2 - x_4 \\
\text{s.t.} \quad & x_1 + x_2 + x_3 + 0x_4 = 2 \\
& 2x_1 + 0x_2 + x_3 + x_4 = 3 \\
& x \geq 0,
\end{aligned}
$$

$$
A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{pmatrix}
$$

$x = (1, 1, 0, 1)^\top$ is an optimal solution but it is not a basic solution. Here $B = \{1, 2, 4\}$. Since A_1, A_2, A_4 are linear dependent, there exists α_i such that $\sum_{i \in B} \alpha_i A_i = 0$,a. We can find $\alpha_1 = 1, \alpha_2 = -1, \alpha_4 = -2$. So $\alpha = (1, -1, 0, -2)^\top$. Let $\epsilon = 1$, then $\bar{x} = x - \epsilon\alpha = (0, 2, 0, 3)^\top$ is an optimal basic feasible solution.

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